Series QSS4R/4

Set – 1

प्रश्न-पत्र कोड
Q.P. Code **65/4/1**

अनुक्रमांक
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- *कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं ।*
- *कृपया जाँच कर लें कि इस प्रश्न-पत्र में **38** प्रश्न हैं ।*
- *प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।*
- ***कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।***
- *इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।*
- *Please check that this question paper contains **23** printed pages.*
- *Please check that this question paper contains **38** questions.*
- *Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.*
- ***Please write down the serial number of the question in the answer-book before attempting it.***
- *15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.*

गणित
MATHEMATICS

*निर्धारित समय : **3** घण्टे*
*Time allowed : **3** hours*

*अधिकतम अंक : **80***
*Maximum Marks : **80***

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This Question Paper contains* ***38*** *questions.* ***All*** *questions are* ***compulsory****.*

(ii) *Question Paper is divided into* ***five*** *Sections – Section* ***A, B, C, D*** *and* ***E****.*

(iii) *In* ***Section A*** *– Questions no.* ***1*** *to* ***18*** *are Multiple Choice Questions (MCQs) and Questions no.* ***19 & 20*** *are Assertion-Reason based questions of* ***1*** *mark each.*

(iv) *In* ***Section B*** *– Questions no.* ***21*** *to* ***25*** *are Very Short Answer (VSA) type questions, carrying* ***2*** *marks each.*

(v) *In* ***Section C*** *– Questions no.* ***26*** *to* ***31*** *are Short Answer (SA) type questions, carrying* ***3*** *marks each.*

(vi) *In* ***Section D*** *– Questions no.* ***32*** *to* ***35*** *are Long Answer (LA) type questions, carrying* ***5*** *marks each.*

(vii) *In* ***Section E*** *– Questions no.* ***36*** *to* ***38*** *are case study based questions, carrying* ***4*** *marks each.*

(viii) *There is no overall choice. However, an internal choice has been provided in* ***2*** *questions in Section* ***B****,* ***3*** *questions in Section* ***C****,* ***3*** *questions in Section* ***D*** *and* ***2*** *questions in Section* ***E****.*

(ix) *Use of calculators is* ***not*** *allowed.*

SECTION – A

This section consists of **20** multiple choice questions of **1** mark each. **20 × 1 = 20**

1. If $\begin{bmatrix} a & c & 0 \\ b & d & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is a scalar matrix, then the value of a + 2b + 3c + 4d is :

(A) 0 (B) 5

(C) 10 (D) 25

2. Given that $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$, matrix A is :

(A) $7\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

(C) $\frac{1}{7}\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

(D) $\frac{1}{49}\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

3. If $A = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$, then the value of $I - A + A^2 - A^3 + ...$ is :

(A) $\begin{bmatrix} -1 & -1 \\ 4 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

4. If $A = \begin{bmatrix} -2 & 0 & 0 \\ 1 & 2 & 3 \\ 5 & 1 & -1 \end{bmatrix}$, then the value of $|\, A\,(\text{adj. } A)\, |$ is :

(A) 100 I

(B) 10 I

(C) 10

(D) 1000

5. Given that $[1 \quad x]\begin{bmatrix} 4 & 0 \\ -2 & 0 \end{bmatrix} = 0$, the value of x is :

(A) –4

(B) –2

(C) 2

(D) 4

6. Derivative of e^{2x} with respect to e^x, is :

(A) e^x

(B) $2e^x$

(C) $2e^{2x}$

(D) $2e^{3x}$

7. For what value of k, the function given below is continuous at $x = 0$?

$$f(x) = \begin{cases} \frac{\sqrt{4+x}-2}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$$

(A) 0 (B) $\frac{1}{4}$

(C) 1 (D) 4

8. The value of $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$ is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{18}$

9. The general solution of the differential equation $x\,dy + y\,dx = 0$ is :

(A) $xy = c$ (B) $x + y = c$

(C) $x^2 + y^2 = c^2$ (D) $\log y = \log x + c$

10. The integrating factor of the differential equation $(x + 2y^2)\frac{dy}{dx} = y \ (y > 0)$ is :

(A) $\frac{1}{x}$ (B) x

(C) y (D) $\frac{1}{y}$

11. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{3}$, then the angle between $2\vec{a}$ and $-\vec{b}$ is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) $\frac{5\pi}{6}$ (D) $\frac{11\pi}{6}$

12. The vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{c} = -3\hat{i} + 4\hat{j} + 4\hat{k}$ represents the sides of

(A) an equilateral triangle

(B) an obtuse-angled triangle

(C) an isosceles triangle

(D) a right-angled triangle

13. Let $\vec{a}$ be any vector such that $|\vec{a}| = a$. The value of

$|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$ is :

(A) a^2 (B) $2a^2$

(C) $3a^2$ (D) 0

14. The vector equation of a line passing through the point (1, –1, 0) and parallel to Y-axis is :

(A) $\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} - \hat{j})$ (B) $\vec{r} = \hat{i} - \hat{j} + \lambda\hat{j}$

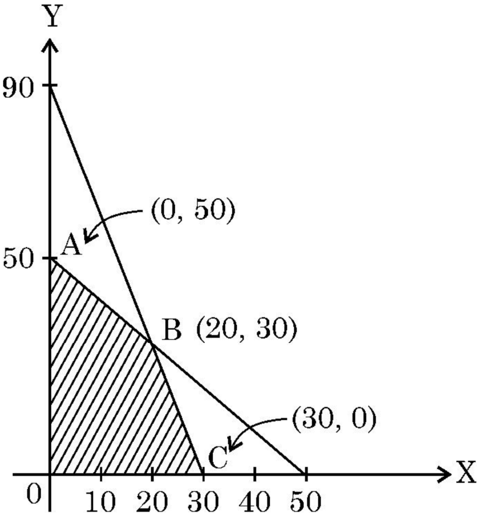
(C) $\vec{r} = \hat{i} - \hat{j} + \lambda\hat{k}$ (D) $\vec{r} = \lambda\hat{j}$

15. The lines $\frac{1-x}{2} = \frac{y-1}{3} = \frac{z}{1}$ and $\frac{2x-3}{2p} = \frac{y}{-1} = \frac{z-4}{7}$ are perpendicular to each other for p equal to :

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$

(C) 2 (D) 3

16. The maximum value of $Z = 4x + y$ for a L.P.P. whose feasible region is given below is :



(A) 50 (B) 110

(C) 120 (D) 170

17. The probability distribution of a random variable X is :

X	0	1	2	3	4
P(X)	0.1	k	2k	k	0.1

where k is some unknown constant.

The probability that the random variable X takes the value 2 is :

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$

(C) $\frac{4}{5}$ (D) 1

18. The function $f(x) = kx - \sin x$ is strictly increasing for

(A) $k > 1$ (B) $k < 1$

(C) $k > -1$ (D) $k < -1$

ASSERTION-REASON BASED QUESTIONS

Questions No. **19** & **20**, are Assertion (A) and Reason (R) based questions carrying **1** mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R).

Select the correct answer from the codes (A), (B), (C) and (D) as given below :

(A) Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of the Assertion (A).

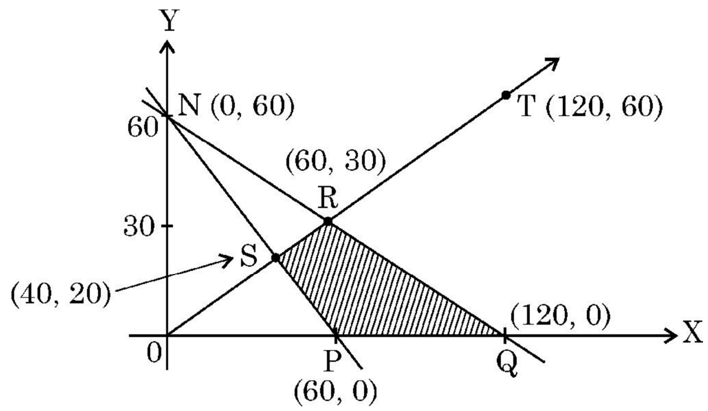
(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** The relation $R = \{(x, y) : (x + y) \text{ is a prime number and } x, y \in N\}$ is not a reflexive relation.

Reason (R) : The number '2n' is composite for all natural numbers n.

20. **Assertion (A) :** The corner points of the bounded feasible region of a L.P.P. are shown below. The maximum value of $Z = x + 2y$ occurs at infinite points.



Reason (R) : The optimal solution of a LPP having bounded feasible region must occur at corner points.

SECTION – B

In this section there are **5** very short answer type questions of **2** marks each.

21. (a) Express $\tan^{-1}\left(\dfrac{\cos x}{1-\sin x}\right)$, where $\dfrac{-\pi}{2} < x < \dfrac{\pi}{2}$ in the simplest form.

OR

(b) Find the principal value of $\tan^{-1}(1) + \cos^{-1}\left(-\dfrac{1}{2}\right) + \sin^{-1}\left(-\dfrac{1}{\sqrt{2}}\right)$.

22. (a) If $y = \cos^3(\sec^2 2t)$, find $\dfrac{dy}{dt}$.

OR

(b) If $x^y = e^{x-y}$, prove that $\dfrac{dy}{dx} = \dfrac{\log x}{(1+\log x)^2}$.

23. Find the interval in which the function $f(x) = x^4 - 4x^3 + 10$ is strictly decreasing.

24. The volume of a cube is increasing at the rate of 6 cm^3/s. How fast is the surface area of cube increasing, when the length of an edge is 8 cm ?

25. Find : $\displaystyle\int \frac{1}{x(x^2-1)}\,dx$.

SECTION – C

In this section there are **6** short answer type questions of **3** marks each.

26. Given that $y = (\sin x)^x \cdot x^{\sin x} + a^x$, find $\dfrac{dy}{dx}$.

27. (a) Evaluate : $\displaystyle\int_0^{\frac{\pi}{4}} \frac{x\,dx}{1+\cos 2x+\sin 2x}$

OR

(b) Find : $\displaystyle\int e^x \left[\frac{1}{(1+x^2)^{\frac{3}{2}}} + \frac{x}{\sqrt{1+x^2}}\right] dx$

28. Find : $\int \frac{3x+5}{\sqrt{x^2+2x+4}}\,dx$

29. (a) Find the particular solution of the differential equation $\frac{dy}{dx} = y \cot 2x$, given that $y\left(\frac{\pi}{4}\right) = 2$.

OR

(b) Find the particular solution of the differential equation

$$(xe^{\frac{y}{x}} + y)\,dx = x\,dy, \text{ given that } y = 1 \text{ when } x = 1.$$

30. Solve the following linear programming problem graphically :

Maximise $Z = 2x + 3y$

subject to the constraints :

$x + y \leq 6$

$x \geq 2$

$y \leq 3$

$x, y \geq 0$

31. (a) A card from a well shuffled deck of 52 playing cards is lost. From the remaining cards of the pack, a card is drawn at random and is found to be a King. Find the probability of the lost card being a King.

OR

(b) A biased die is twice as likely to show an even number as an odd number. If such a die is thrown twice, find the probability distribution of the number of sixes. Also, find the mean of the distribution.

SECTION – D

In the section there are **4** long answer type questions of **5** marks each.

32. (a) Sketch the graph of $y = x|x|$ and hence find the area bounded by this curve, X-axis and the ordinates $x = -2$ and $x = 2$, using integration.

OR

(b) Using integration, find the area bounded by the ellipse $9x^2 + 25y^2 = 225$, the lines $x = -2$, $x = 2$, and the X-axis.

33. (a) Let $A = R - \{5\}$ and $B = R - \{1\}$. Consider the function $f : A \to B$, defined by $f(x) = \dfrac{x-3}{x-5}$. Show that f is one-one and onto.

OR

(b) Check whether the relation S in the set of real numbers R defined by $S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.

34. If $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, find A^{-1} and hence solve the following system of equations :

$2x + y - 3z = 13$

$3x + 2y + z = 4$

$x + 2y - z = 8$

35. (a) Find the distance between the line $\dfrac{x}{2} = \dfrac{2y-6}{4} = \dfrac{1-z}{-1}$ and another line parallel to it passing through the point $(4, 0, -5)$.

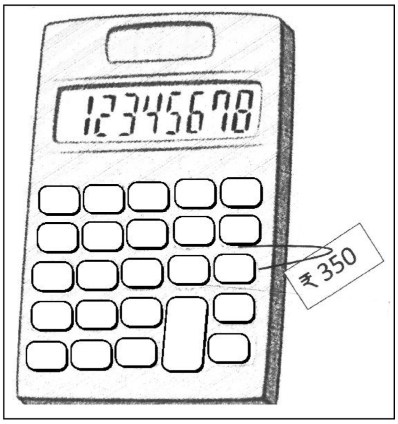
OR

(b) If the lines $\dfrac{x-1}{-3} = \dfrac{y-2}{2k} = \dfrac{z-3}{2}$ and $\dfrac{x-1}{3k} = \dfrac{y-1}{1} = \dfrac{z-6}{-7}$ are perpendicular to each other, find the value of k and hence write the vector equation of a line perpendicular to these two lines and passing through the point $(3, -4, 7)$.

SECTION – E

In this section, there are **3** case study based questions of **4** marks each.

36. A store has been selling calculators at ₹ 350 each. A market survey indicates that a reduction in price (p) of calculator increases the number of units (x) sold. The relation between the price and quantity sold is given by the demand function $p = 450 - \frac{1}{2}x$.

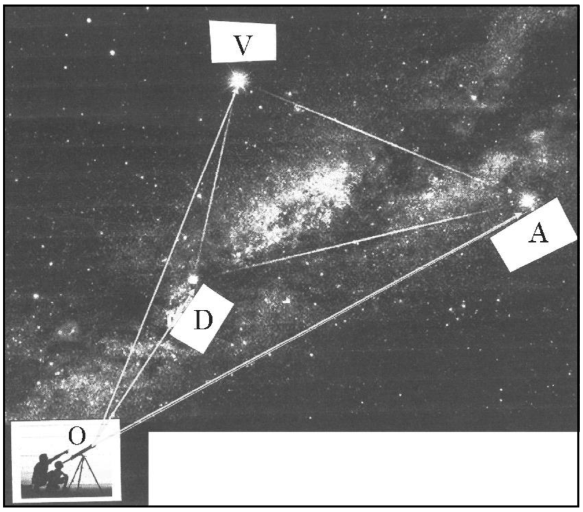


Based on the above information, answer the following questions :

(i) Determine the number of units (x) that should be sold to maximise the revenue $R(x) = x p(x)$. Also, verify the result.

(ii) What rebate in price of calculator should the store give to maximise the revenue ?

37. An instructor at the astronomical centre shows three among the brightest stars in a particular constellation. Assume that the telescope is located at O(0, 0, 0) and the three stars have their locations at the points D, A and V having position vectors $2\hat{i} + 3\hat{j} + 4\hat{k}$, $7\hat{i} + 5\hat{j} + 8\hat{k}$ and $-3\hat{i} + 7\hat{j} + 11\hat{k}$ respectively.



Based on the above information, answer the following questions :

(i) How far is the star V from star A ? **1**

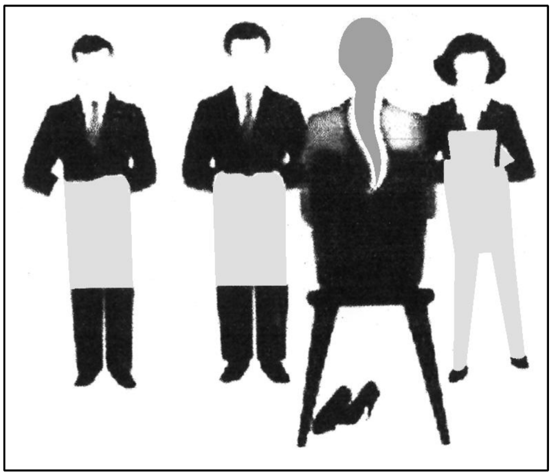
(ii) Find a unit vector in the direction of $\overrightarrow{DA}$. **1**

(iii) Find the measure of $\angle VDA$. **2**

OR

(iii) What is the projection of vector $\overrightarrow{DV}$ on vector $\overrightarrow{DA}$? **2**

38. Rohit, Jaspreet and Alia appeared for an interview for three vacancies in the same post. The probability of Rohit's selection is $\frac{1}{5}$, Jaspreet's selection is $\frac{1}{3}$ and Alia's selection is $\frac{1}{4}$. The event of selection is independent of each other.



Based on the above information, answer the following questions :

(i) What is the probability that at least one of them is selected ? **1**

(ii) Find $P(G \mid \overline{H})$ where G is the event of Jaspreet's selection and $\overline{H}$ denotes the event that Rohit is not selected. **1**

(iii) Find the probability that exactly one of them is selected. **2**

OR

(iii) Find the probability that exactly two of them are selected. **2**